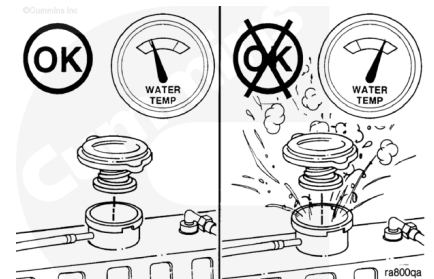


Trainings ▾

Drain

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



The QSK45 and QSK60 industrial engines use a two-pump/two-loop cooling system. The primary radiator provides cooling for the engine using the water pump mounted on the right-bank side of the engine. An LTA water pump is mounted on the left-bank side of the engine on the front gear cover. The LTA water pump circulates coolant through the two-pass core of the LTA radiator and back through the aftercoolers. The two center thermostats control temperature as the coolant passes through the thermostat housing.

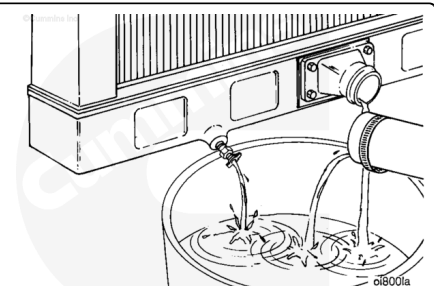
Remove the radiator cap after the engine is cool.

Open the petcocks on each turbocharger.

Aftercooled Engines

⚠ WARNING ⚠

Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.



This cooling system uses the same radiator to cool the aftercoolers as is used to cool the engine. Refer to Procedure 200-003 in Section F.

(/qs3/pubsys2/xml/en/procedures/56/56-200-003.html)

Open the draincock at the bottom of the radiator.

Remove the lower radiator hose.

Drain the cooling system.

Do **not** allow the cooling system to dry out.

LTA

⚠ WARNING ⚠

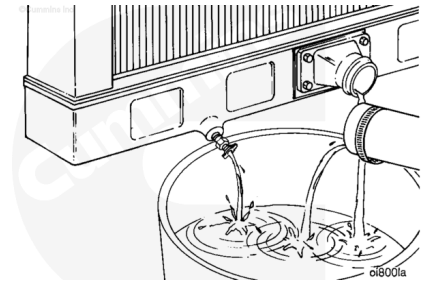
Coolant is toxic. Keep away from children and pets. If not reused, dispose of in accordance with local environmental regulations.

This system cools the aftercoolers, using a radiator separate from that used to cool the engine. Refer to Procedure 200-003 in Section F. (</qs3/pubsys2/xml/en/procedures/56/56-200-003.html>)

Drain the coolant.

The low-temperature aftercooling (LTA) cooling system is drained by opening the draincock in the LTA radiator or disconnecting the piping.

Do **not** allow the cooling system to dry out.

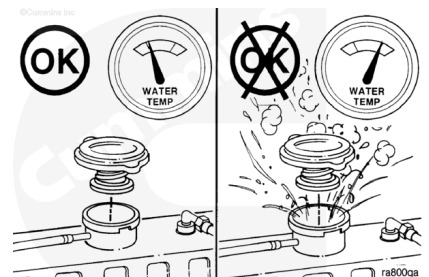


Flush

⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.

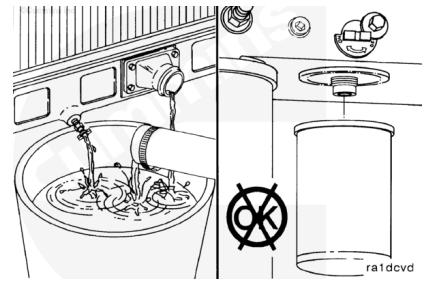
Drain the cooling system.



The performance of Restore™ depends on time, temperature, and concentration levels. An extremely scaled or flow-restricted system, for example, can require higher concentrations of cleaners, higher temperatures, or longer cleaning time. Restore™ can be safely used up to twice the recommended concentration levels. Extremely scaled or fouled systems can require more than one cleaning.



Do **not** remove the coolant filters.



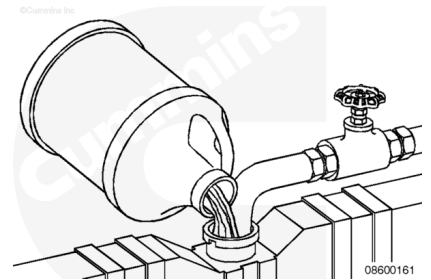
⚠ CAUTION ⚠

Fleetguard® Restore™ contains no antifreeze. Do not allow the cooling system to freeze during the cleaning operation.

Close all petcocks to avoid air entrapment in the cooling system.

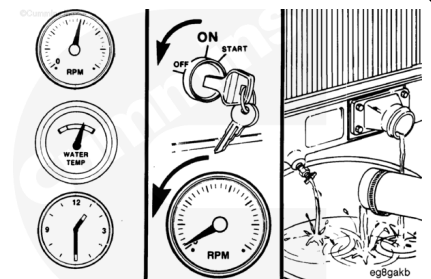
Immediately add 3.8 liters [1 gal] of Fleetguard® Restore™, or equivalent, for each 38 to 57 liters [10 to 15 gal] of cooling system capacity, and fill the system with plain water.

Turn the heater temperature switch to HIGH to allow maximum coolant flow through the heater core. The blower does **not** have to be on.

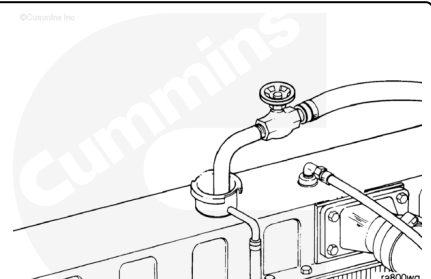


Operate the engine at a coolant temperature of at least 85°C [185°F] for 1 to 1-½ hours.

Shut the engine OFF, allow to cool to 50°C [120°F], and drain the cooling system.



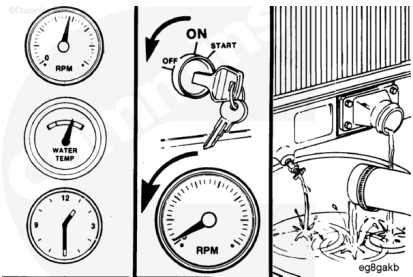
Fill the cooling system with clean water.



Operate the engine at high idle for 5 minutes with the coolant temperature above 85°C [185°F].

Shut the engine OFF, allow to cool to 50°C [120°F], and drain the cooling system.

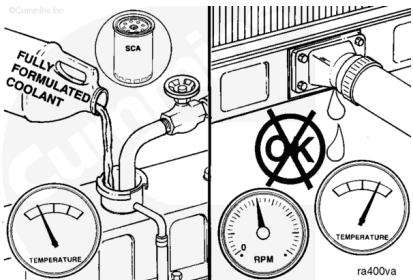
If the water drained is still dirty, the system **must** be flushed again until the water is clean.



Install a new coolant filter.

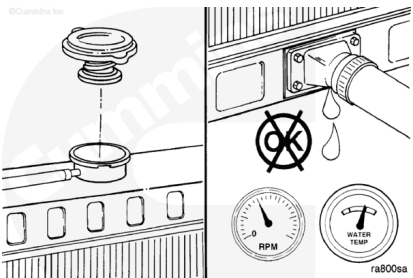
Fill the cooling system with fully formulated coolant.

Use additional SCA to bring the coolant to the correct SCA concentration level. Refer to Procedure 018-018 in Section V. (/qs3/pubsys2/xml/en/procedures/56/56-018-018.html)



Install the pressure cap. Operate the engine until the coolant reaches a temperature of 70°C [158°F] and check for coolant leaks.

Measurements		
	kpa	psi
Minimum Cooling System Pressure Cap Capacity	76	11



Pressure Test

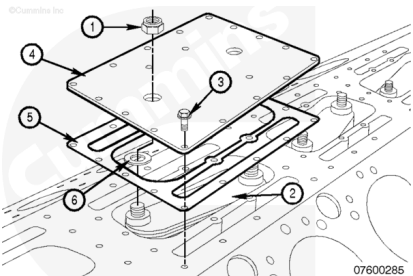
Internal

Check the coolant level and fill if necessary.

To aid in determining the location of the coolant leak, it could be necessary to add fluorescent tracer, Part Number 3377438, to the coolant.

Add one unit of fluorescent tracer to each 38 liters [10 gal] of coolant.

Idle the engine for 5 to 10 minutes or until normal operating temperature is reached to allow the dye to circulate through the cooling system.



Use a high intensity black light, Part Number 3163337, or equivalent, to illuminate the dye.

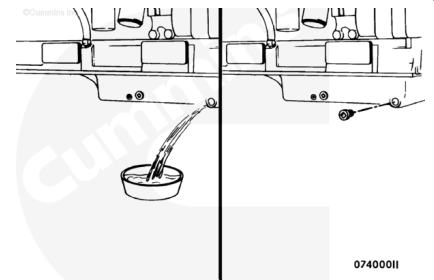
Contact a Cummins® Authorized Repair Location if the customer experiences a coolant loss that can **not** be attributed to an external coolant leak or if coolant being pushed out of the overflow.

Check the oil cooler covers for coolant leakage. Remove the cooler covers, and perform an inspection and pressure test as required. Refer to Procedure 007-045 in Section 7.

(/qs3/pubsys2/xml/en/procedures/56/56-007-045-tr.html)

⚠ WARNING ⚠

Some state and federal agencies have determined that used engine oil can be carcinogenic and cause reproductive toxicity. Avoid inhalation of vapors, ingestion, and prolonged contact with used engine oil. If not reused, dispose of in accordance with local environmental regulations.



If the oil cooler cores are **not** leaking, drain the oil. Refer to Procedure 007-037 in Section 7.

(/qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html)

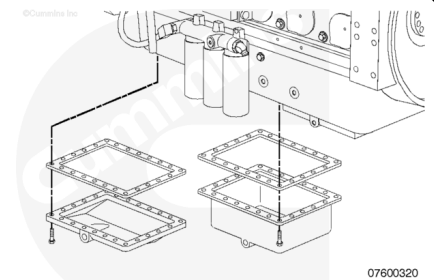
Check for coolant in the oil.

If the oil does **not** contain coolant, check the aftercooler. Refer to Procedure 010-002 in Section 10.

(/qs3/pubsys2/xml/en/procedures/56/56-010-002-tr.html)

⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift this component.

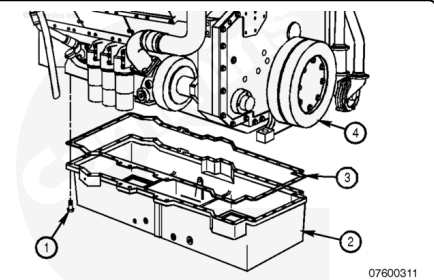


Remove the oil pan. Refer to Procedure 007-025 in Section 7.

(/qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html)

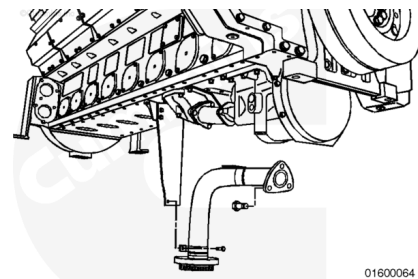
⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift this component.



Remove the oil pan adapter. Refer to Procedure 007-027 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html>)

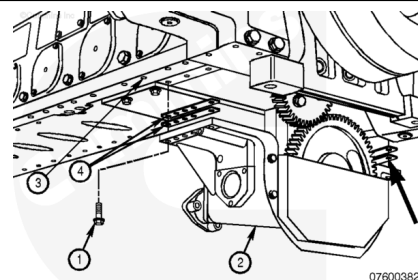
Remove the oil suction and transfer tubes. Refer to Procedure 007-035 (</qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html>) and Refer to Procedure 007-040 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-040-tr.html>)



⚠ WARNING ⚠

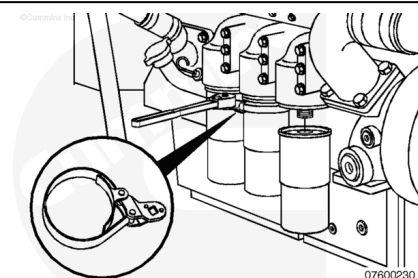
This component weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift this component.

Remove the oil pump Refer to Procedure 007-031 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html>)



Remove the lubricating oil filters. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html>)

Discard the filters if they are **not** required for failure analysis.

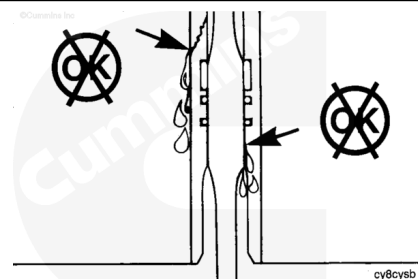


Check for leakage on the inside and outside of the cylinder liner.

If coolant is leaking on the inside of the cylinder liner, remove the cylinder head and pressure test the cooling system before removing the cylinder liner.

Remove all cylinder liners that are leaking externally. Refer to Procedure 001-028 in Section 1. (</qs3/pubsys2/xml/en/procedures/56/56-001-028-tr.html>)

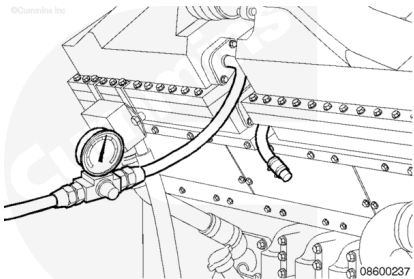
If the cylinder liners are **not** leaking, check the aftercooler element.



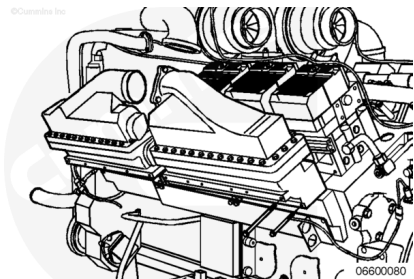
Disconnect the aftercooler coolant lines. Plug one connector, and apply air pressure to the other connector.

Measurements		
	kpa	psi
Air Pressure:	415	60

If the air pressure decreases, replace the aftercooler element
Refer to Procedure 010-008 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-008.html)



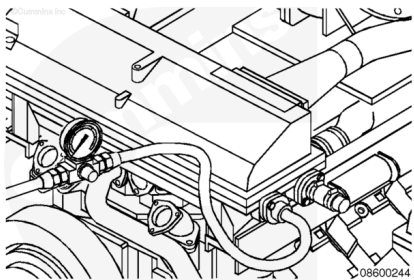
If the aftercooler element is **not** leaking, check the intercooler elements.



Disconnect the intercooler coolant connectors. Plug one connector and apply air pressure to the other connector.

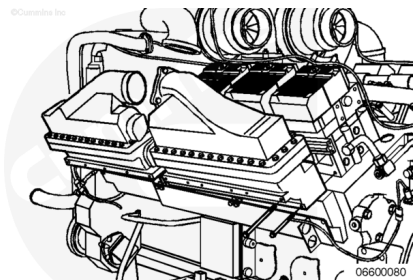
Measurements		
	kpa	psi
Air Pressure:	415	60

If the air pressure decreases, replace the intercooler element.
Refer to Procedure 010-084 in Section 10.
(/qs3/pubsys2/xml/en/procedures/56/56-010-084.html)



If the intercooler element is **not** leaking, check the cylinder heads.

Inspect the cylinder heads. Refer to Procedure 002-004 in Section 2. (/qs3/pubsys2/xml/en/procedures/56/56-002-004-tr.html)



Check the cylinder head intake ports and exhaust ports for coolant/fluorescent dye using a black light. Presence of fluorescent dye indicates a crack.

Note : A green glow would be seen in the area where the leak has occurred. It is the coolant with tracer dye

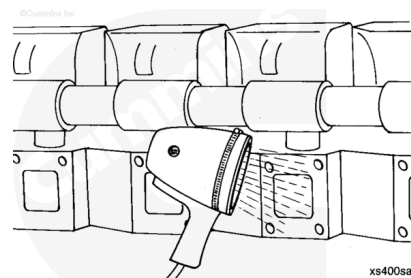
glowing from the light of the black light.

Remove the valve covers. Shine the black light in the overhead around the valve springs. If a spring deck crack is present and leaking coolant, it will glow under the black light.

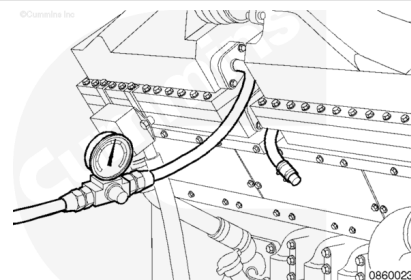
Not all cracks will easily be detected by this procedure. It is also possible to detect signs of coolant in the exhaust by looking for signs of either rust or evidence of steam cleaning in either the exhaust port in the head or its corresponding position in the exhaust manifold.

If coolant or fluorescent dye is found in the intake/exhaust ports, remove the cylinder head and test for defects. Replace if defective. Refer to Procedure 002-004 in Section 2.

(/qs3/pubsys2/xml/en/procedures/56/56-002-004-tr.html)

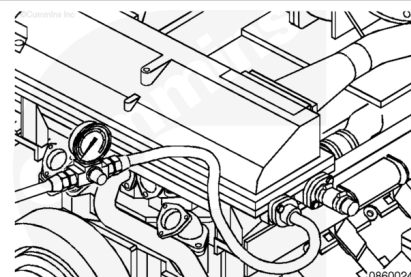


Remove the test equipment from the aftercooler. Connect coolant lines.



Remove the test equipment from the intercooler. Remove the plug from the outlet tube.

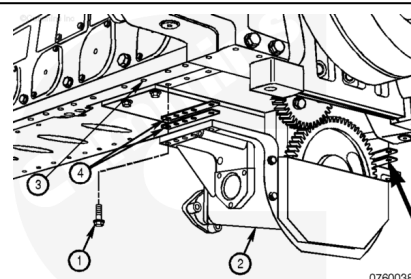
Replace the pressure cap.



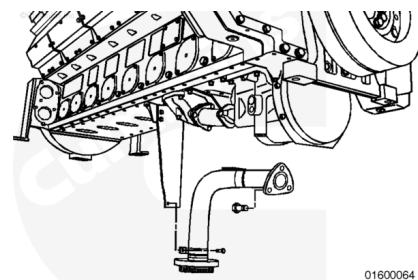
⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift this component.

Install the lubricating oil pump. Refer to Procedure 007-031 in Section 7. (/qs3/pubsys2/xml/en/procedures/56/56-007-031-tr.html)



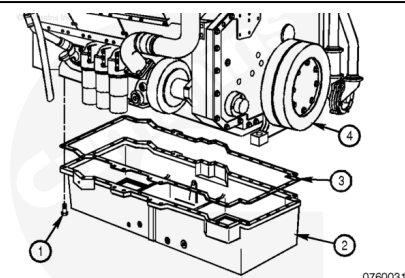
Install the oil transfer and suction tubes. Refer to Procedure 007-035 (</qs3/pubsys2/xml/en/procedures/56/56-007-035-tr.html>) and Refer to Procedure 007-040 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-040-tr.html>)



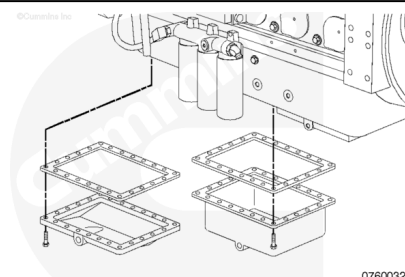
⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To avoid personal injury, use a hoist or get assistance to lift this component.

Install the oil pan adapter. Refer to Procedure 007-027 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-027-tr.html>)

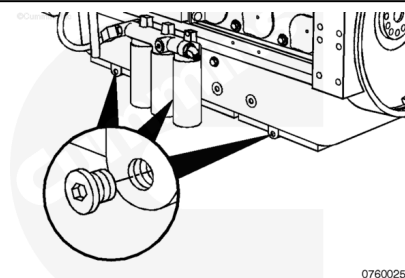


Install the oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-025-tr.html>)
Install the oil pan adapter cover plate. Refer to Procedure 007-026 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-026-tr.html>)

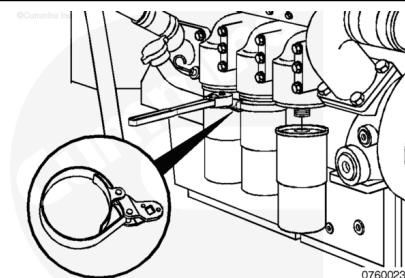


Install and tighten the drain plug(s).

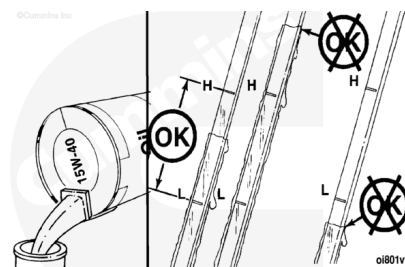
Torque Value: 27 n•m [239 in-lb]



Install the new lubricating oil filters. Refer to Procedure 007-013 in Section 7. (</qs3/pubsys2/xml/en/procedures/56/56-007-013-tr.html>)

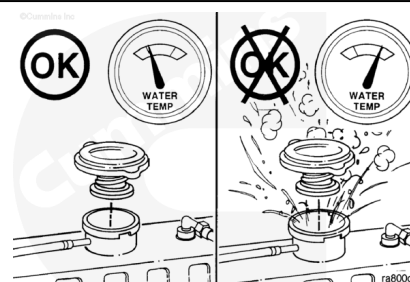


Fill the engine with clean engine oil to the correct level. Refer to Procedure 007-037 in Section 7 (</qs3/pubsys2/xml/en/procedures/56/56-007-037-tr.html>) for the oil capacity of the engine.



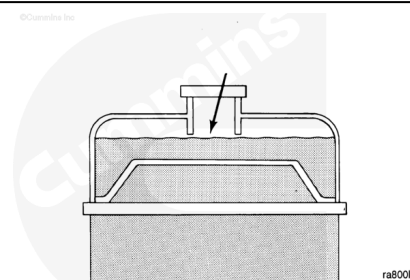
External

To confirm and locate external coolant leaks, it is **not** necessary to warm the engine.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



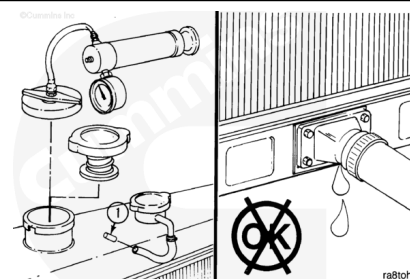
Check the coolant level and fill if necessary.

Add heavy-duty coolant. Refer to Procedure 018-018 in Section V (</qs3/pubsys2/xml/en/procedures/56/56-018-018.html>) for specifications.

To confirm and locate external coolant leaks that occur during normal engine operation, complete this test while the engine is warm.

⚠ CAUTION ⚠

Do not apply more than 140 kPa [20 psi] of air pressure to the cooling system. Excessive air pressure can damage the water pump seal.



If the radiator is equipped with a pressure relief valve, install a plug in the overflow tube (1).

Install the pressure tester on the radiator fill neck or surge tank (if equipped).

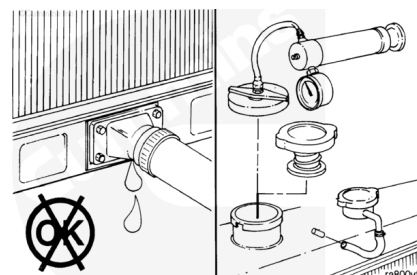
Apply maximum air pressure.

Measurements

	kpa	psi
Maximum Air Pressure:	140	20

Maintain the air pressure for a minimum of 5 minutes. Check for leaks. Any leaks **must** be repaired.

Remove the pressure test equipment and the plug from the pressure relief valve overflow tube (if equipped).

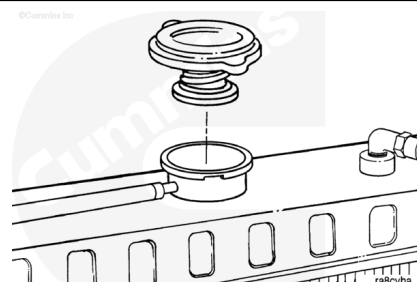


Install the coolant system pressure cap.

Install the lower radiator hose(s).

Tighten the hose clamps.

Torque Value: 5 n·m [44 in-lb]



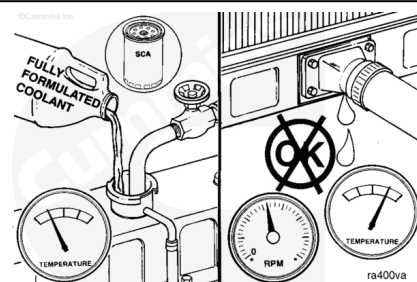
Fill

Make sure the radiator draincocks are closed.

Open the cooling system venting petcocks.

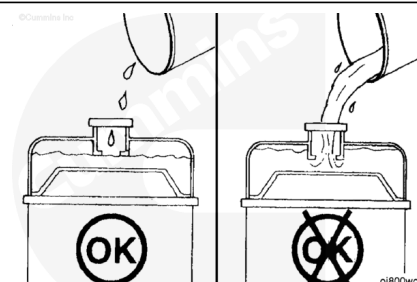
Use fully formulated coolant to fill the cooling system.

Use the correct units of SCA to obtain the correct cooling system protection. Refer to Procedure 018-018 in Section V. (</qs3/pubsys2/xml/en/procedures/56/56-018-018.html>)



⚠ CAUTION ⚠

Fill the cooling system slowly to avoid trapping air in the cooling system. Air trapped in the system can cause water pump cavitation and inadequate coolant level.



Begin filling the cooling system with coolant.

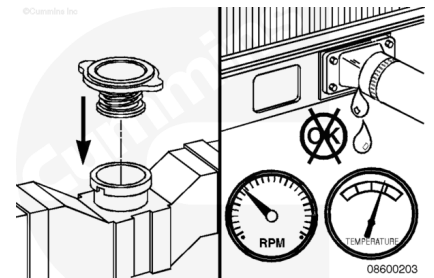
Fill until a continuous stream of coolant begins to flow from each opened petcock in the coolant system.

Close each petcock as the continuous stream of coolant begins to flow from that petcock.

Fill the system with the coolant to the bottom of the radiator fill neck.

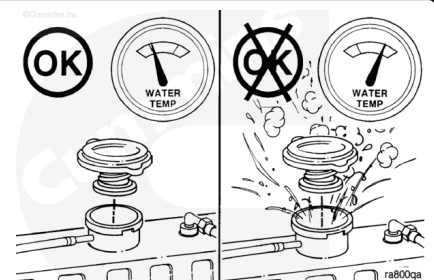
Install the cooling system pressure cap.

Operate the engine until the coolant reaches a temperature of 85°C [185°F]. Check for leaks.



⚠ WARNING ⚠

Do not remove the pressure cap from a hot engine. Wait until the coolant temperature is below 50°C [120°F] before removing the pressure cap. Heated coolant spray or steam can cause personal injury.



Shut the engine off, and allow it to cool.

Check the coolant level.